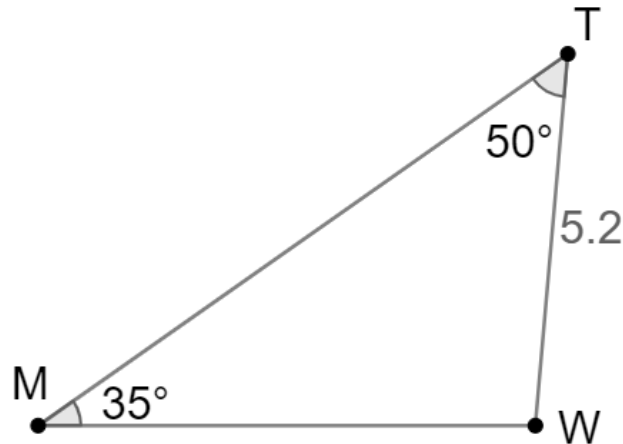


Geometry
Unit 5
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

Triangle TWM is shown where $m\angle M = 35^\circ$, $m\angle T = 50^\circ$, and $WT = 5.2$ units.



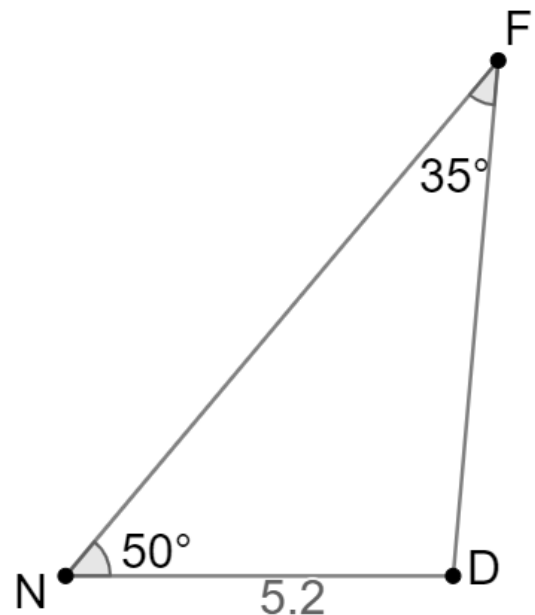
Complete the steps to describe how to construct a triangle that is congruent to $\triangle TWM$ using the angle-angle-side criteria where the first step is to draw a line that is longer than all of the sides of $\triangle TWM$.

1. Use a compass or straightedge to copy TW . Label the endpoints as N and D .
2. Use a compass to copy angle T . Construct a congruent angle using point N as the vertex.
3. Use a compass to copy angle M . Construct a congruent angle so that one side of the angle goes through point D .

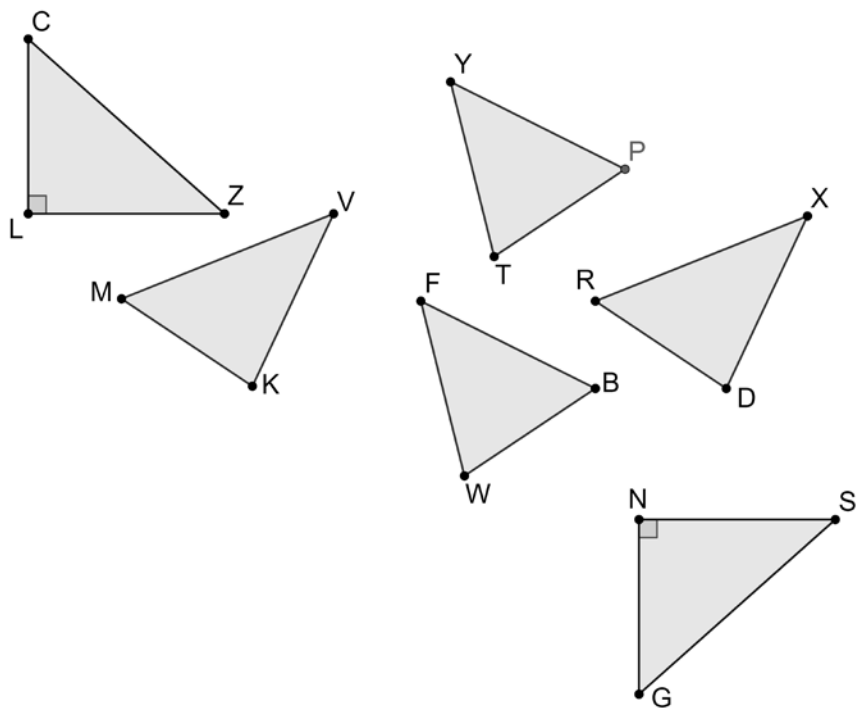
Shanice used the steps to create a triangle that is congruent to $\triangle TWM$. Her triangle is shown.

4. Describe what geometric tools Shanice can use to ensure $\triangle TWM \cong \triangle NDF$.

Shanice can use a ruler, compass, protractor, and/or patty paper.



Triangles CLZ , MKV , YPT , BFW , GSN , and XDR are shown.



Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
5.	$\overline{LZ} \cong \overline{NS}$, $\overline{CZ} \cong \overline{SG}$, $\angle LZC \cong \angle NSG$	Yes	SAS Congruence or HL
6.	$\angle PTY \cong \angle BWF$, $\overline{YP} \cong \overline{FB}$, $\angle TYP \cong \angle WFB$	Yes	AAS Congruence
7.	$\overline{VK} \cong \overline{XD}$, $\overline{MK} \cong \overline{DR}$, $\angle KVM \cong \angle DXR$	No	
8.	$\overline{CZ} \cong \overline{GS}$, $\overline{CL} \cong \overline{GN}$	Yes	HL
9.	$\overline{MV} \cong \overline{XR}$, $\overline{KM} \cong \overline{RD}$, $\overline{XD} \cong \overline{VK}$	Yes	SSS Congruence
10.	$\angle VMK \cong \angle XRD$, $\overline{VK} \cong \overline{XD}$, $\overline{MK} \cong \overline{DR}$	No	